

Arbeitsbesprechung 06.10.2026

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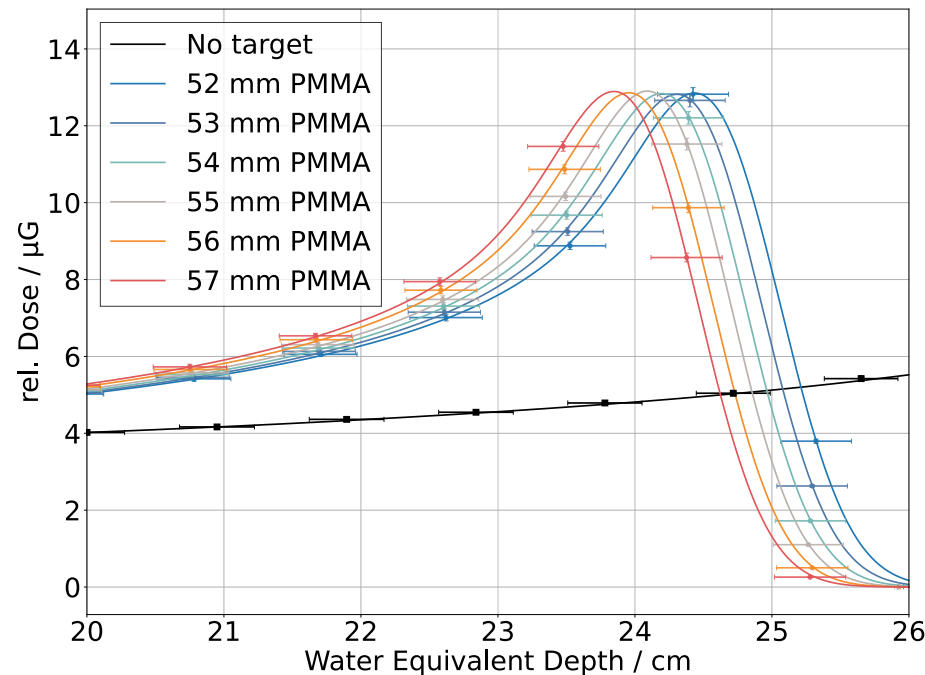
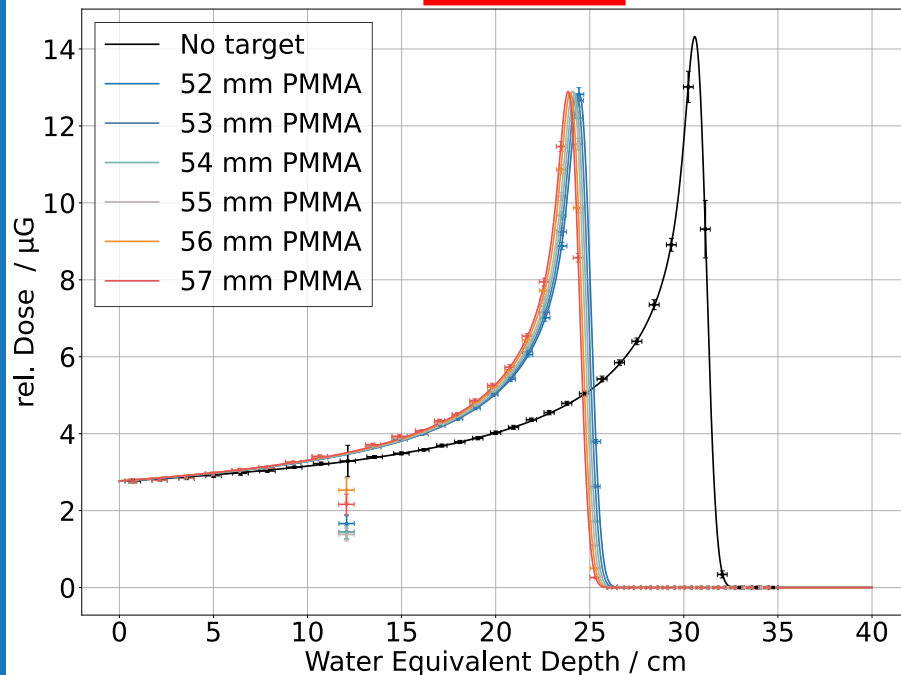
²University of Applied Sciences Giessen, Giessen, Germany

Material Depended Bragg Curve Shape

Phantom	R_0 / cm	σ / cm	t / cm	$t_{\text{nom.}}$ / cm	Δt / μm	$\Delta t_{\%}$ / %
None	30.997 ± 0.002	0.453 ± 0.005	-	-	-	-
52 mm	24.850 ± 0.003	0.449 ± 0.003	6.147 ± 0.004	6.136	110	0.18
53 mm	24.717 ± 0.004	0.445 ± 0.002	6.280 ± 0.004	6.254	260	0.41
54 mm	24.601 ± 0.004	0.441 ± 0.003	6.396 ± 0.004	6.372	240	0.38
55 mm	24.493 ± 0.004	0.435 ± 0.003	6.504 ± 0.004	6.490	140	0.22
56 mm	24.362 ± 0.004	0.441 ± 0.006	6.635 ± 0.004	6.608	270	0.41
57 mm	24.253 ± 0.003	0.440 ± 0.008	6.744 ± 0.003	6.726	180	0.27

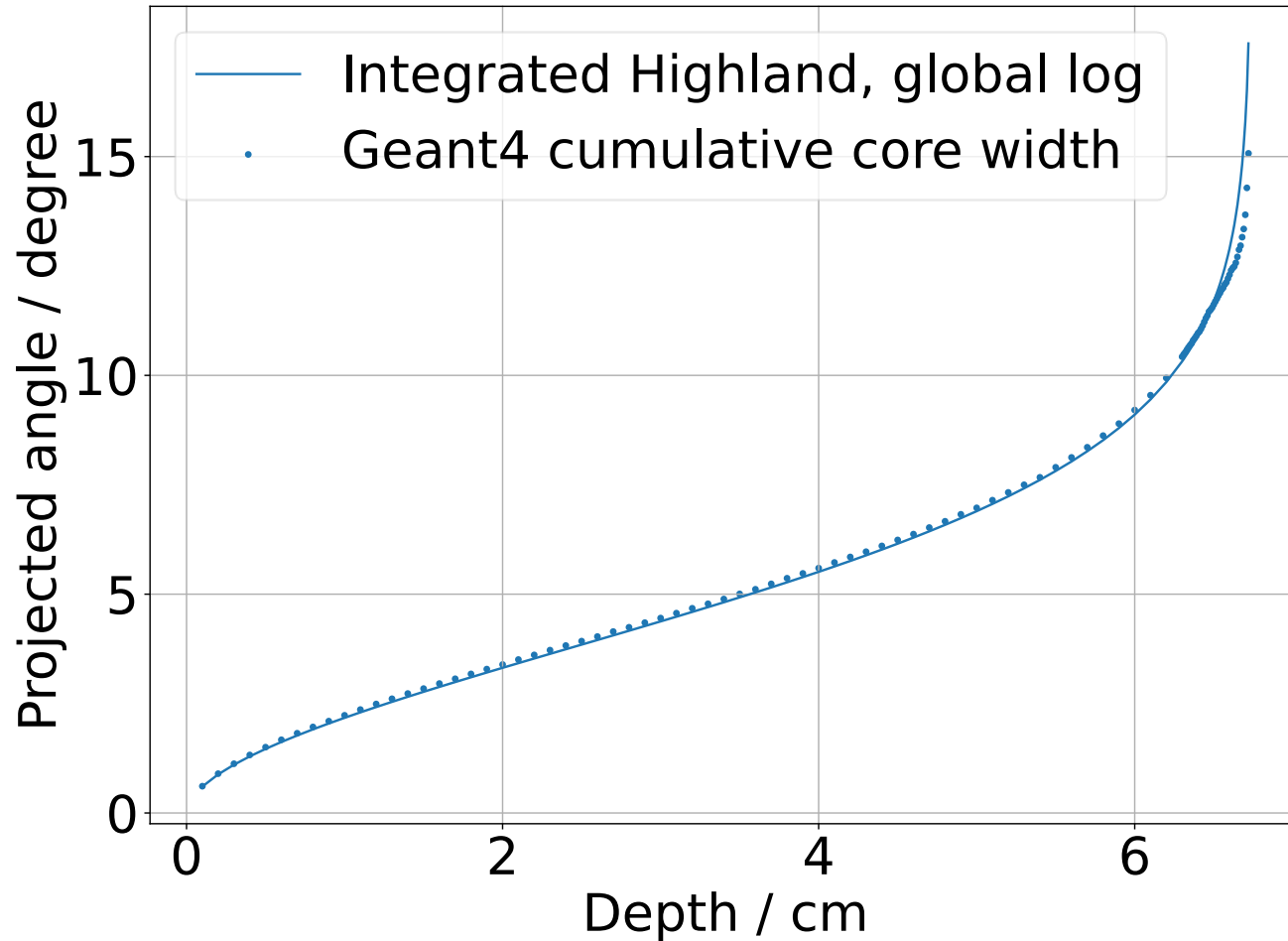


- Material differences:
 - Range straggling
 - Mean scattering angle
 - Inelastic nuclear interactions



Multiple Coulomb Scattering Correction

- Cumulative highland scattering angle



Pivot: Unfolding using PDF/Response Function

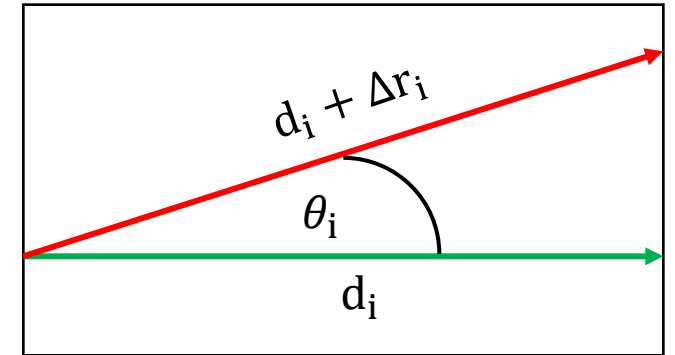
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Pivot: Unfolding using PDF/Response Function

- Correct MCS contribution by unfolding PbWO4 MCS and refolding Water MCS
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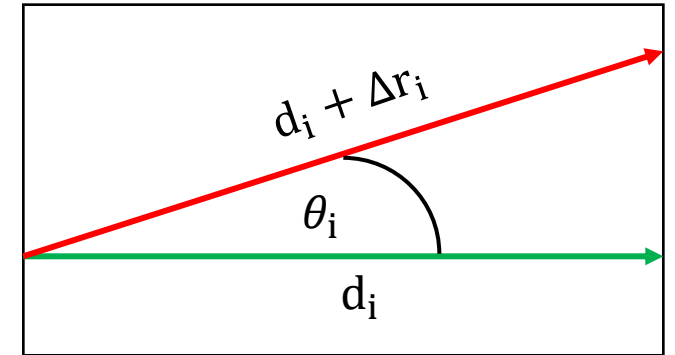
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- Angular layer and cumulative kick known from Highland
 - Calculate path length increase by layers



$$\Delta r_i = \frac{d_i}{\cos(\theta_{i,x})} - d_i + \frac{d_i}{\cos(\theta_{i,y})} - d_i = \frac{d_i}{2}(\theta_{i,x}^2 + \theta_{i,y}^2), \text{ using } \frac{1}{\cos(x)} \approx 1 + \frac{x^2}{2} \quad (2)$$

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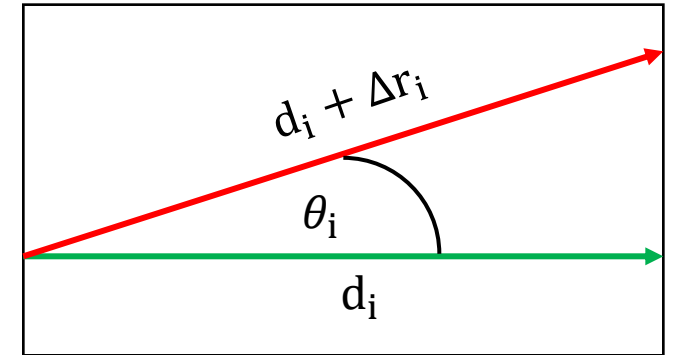
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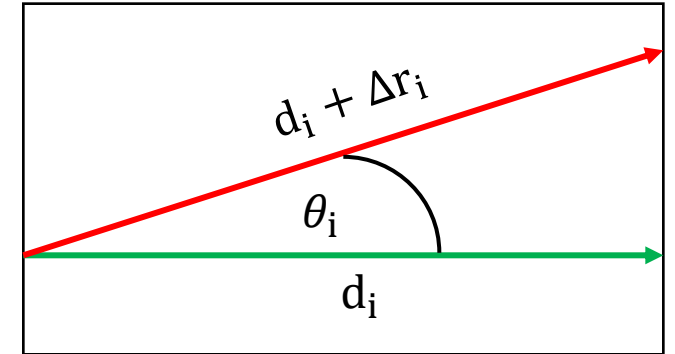
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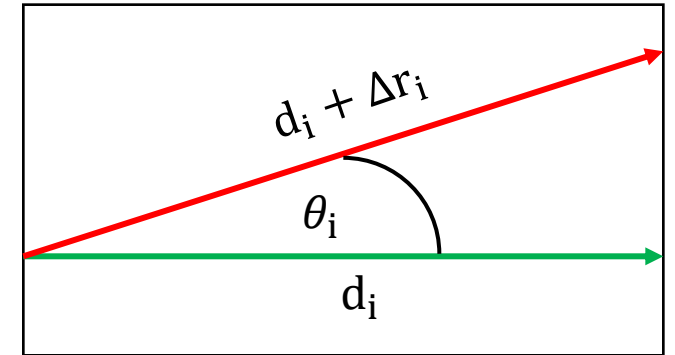
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- Coordinate transformation

$$\Delta R = \sum_i Y_i, \text{ with } f_{Y_i}(y) = \frac{1}{2\lambda_i} \exp\left(-\frac{y}{2\lambda_i}\right) \quad (8)$$

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Final Probability Formula

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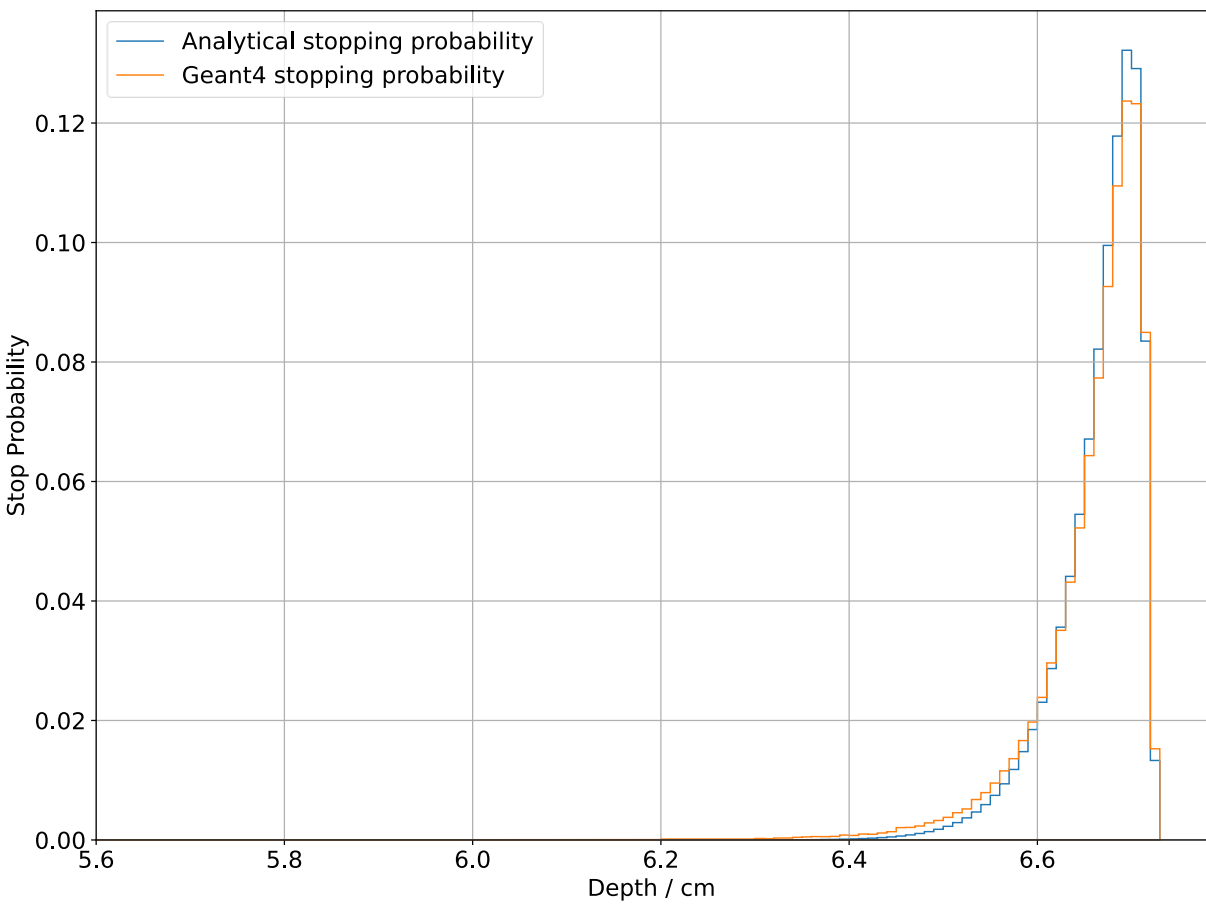
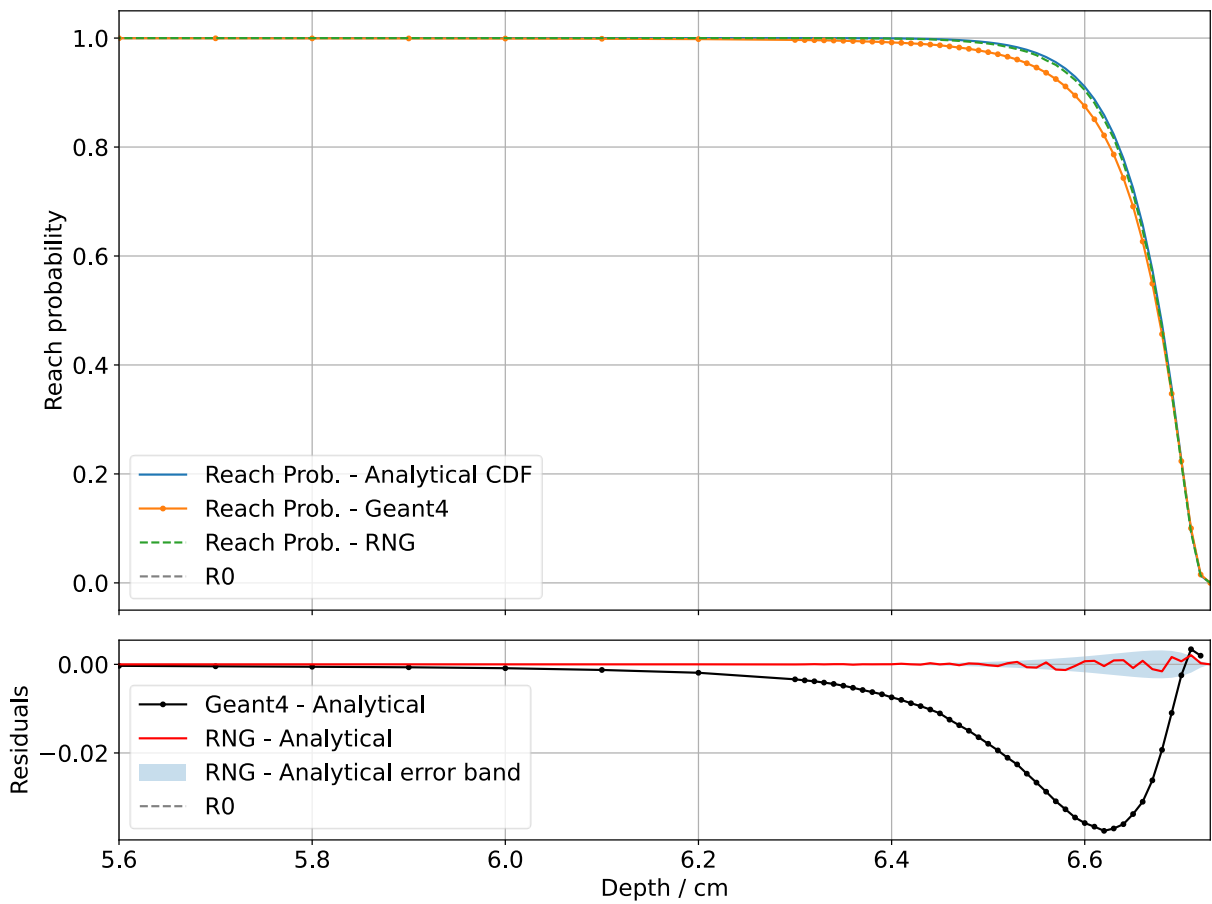
From reach probability we can calculate the stopping probability by partial derivative:

$$P_{\text{stop}}(z) = -\frac{dP_{\text{reach}}(z)}{dz} \quad (16)$$

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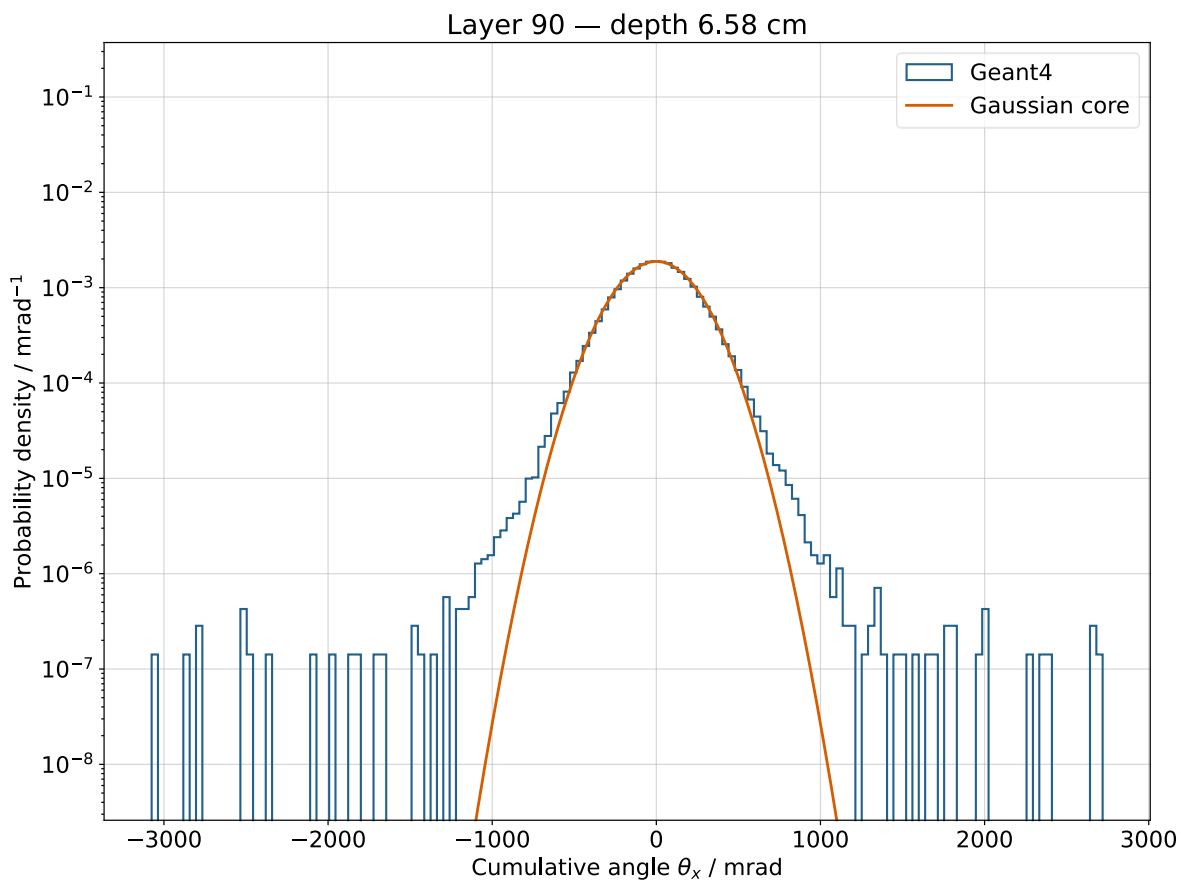
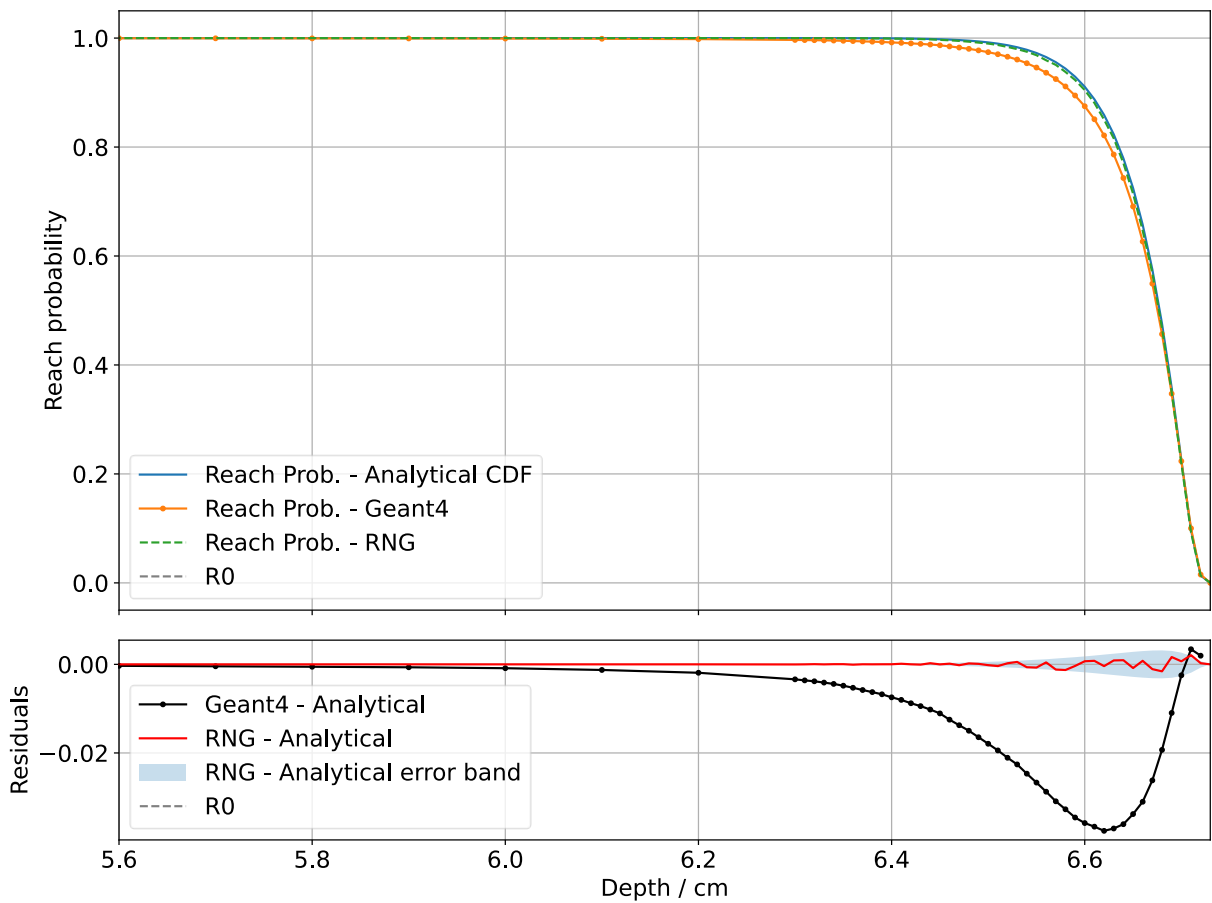
Geant4 Comparision

For PbWO4



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Geant4 Comparision

For H2O

